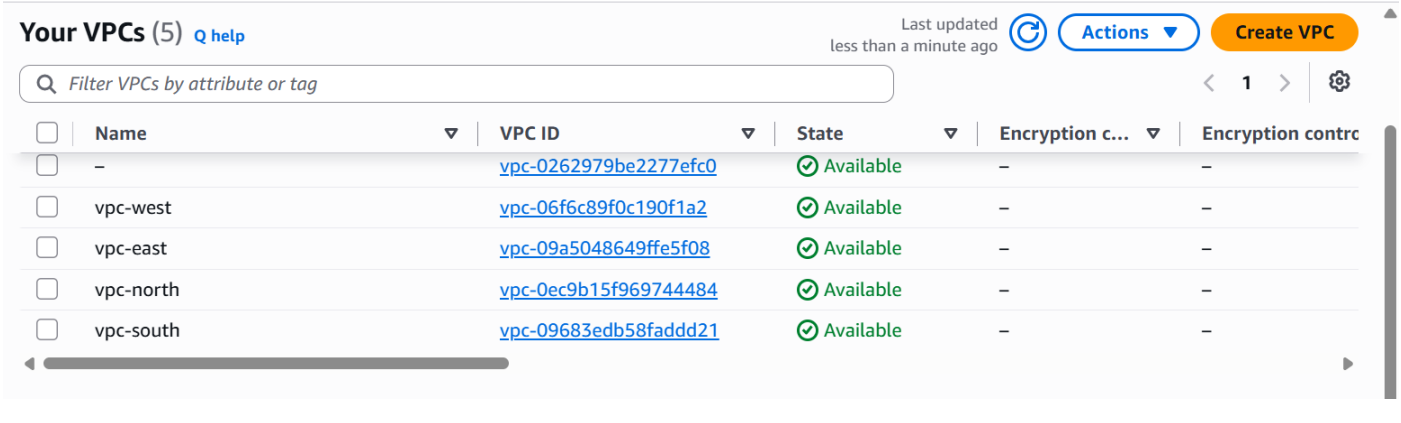
Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.

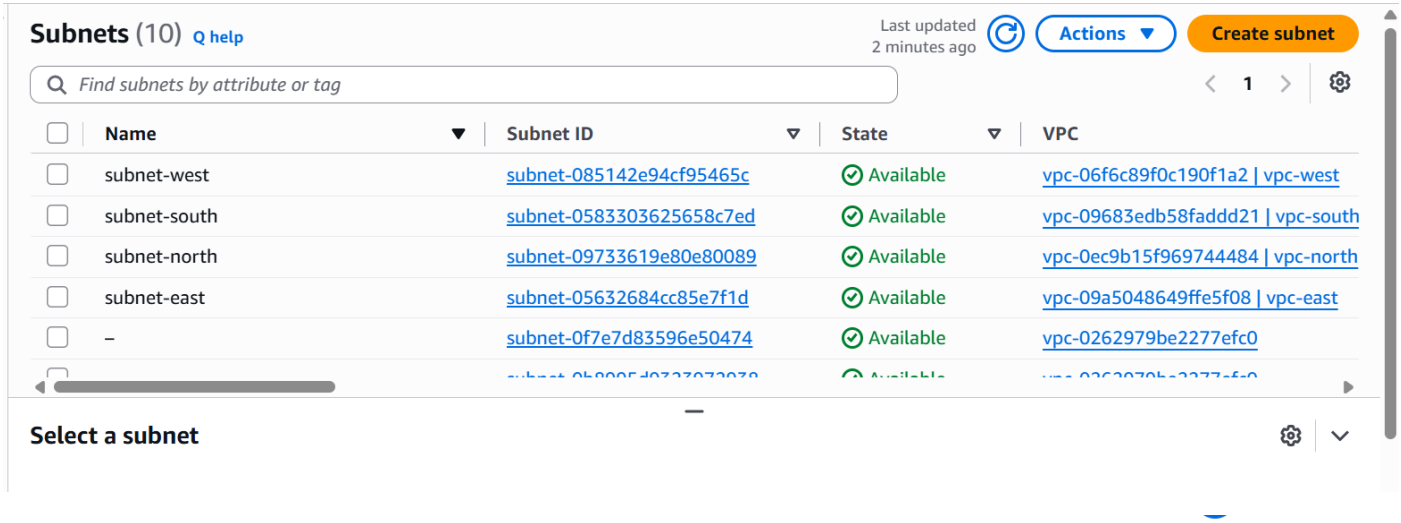
Steps:

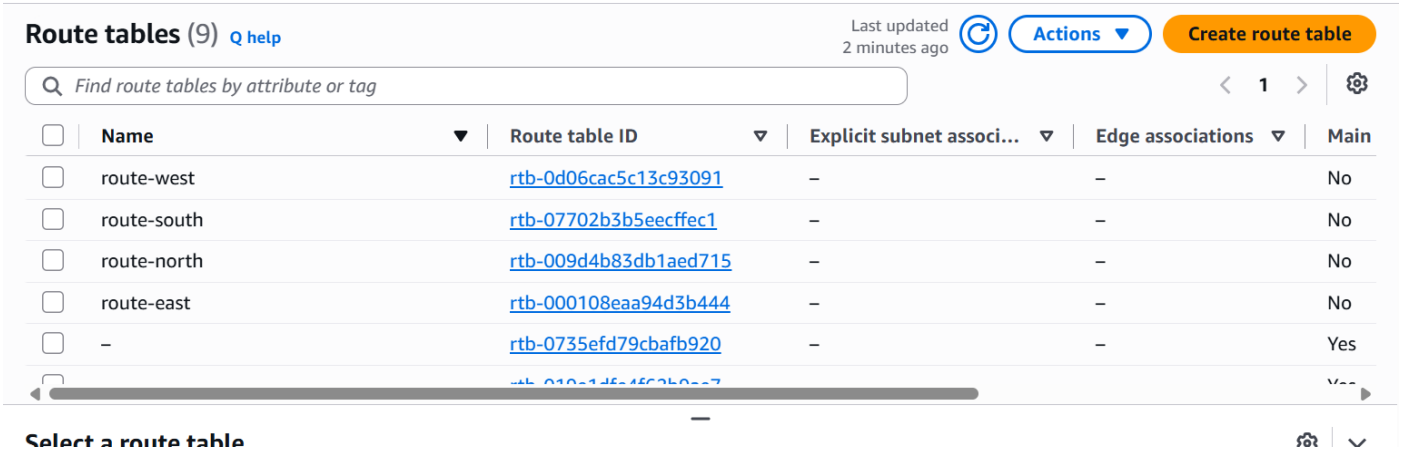
First, I have created 4 VPCs 1 public and 3 privates.
Created 4 subnets for 4 VPCs → Created 4 Routes → 1 Internet gate Way → 1 Nat Gateway
Then Created Transit Gateway → add 4 attachments to Transits Gate Way.
Then Go-to Subnets and edit Routes with IPv4s.

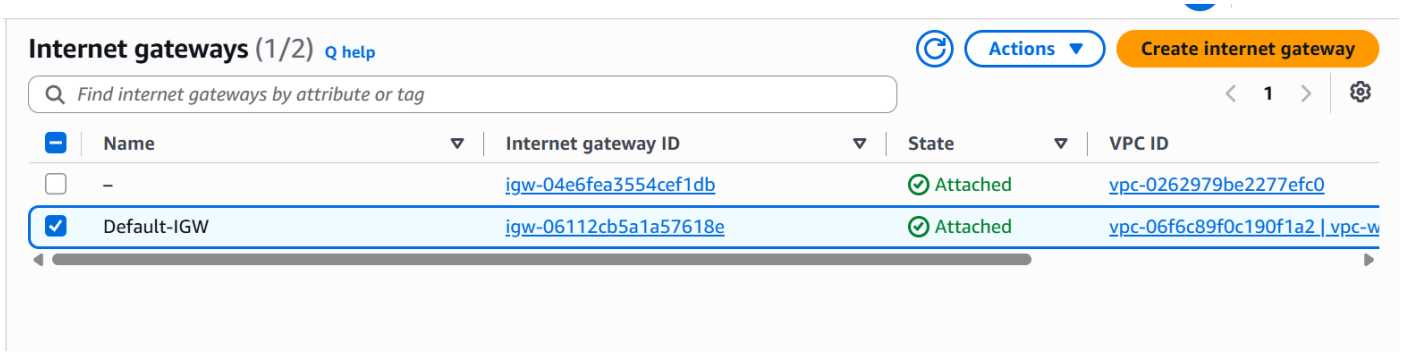
Then Created 4 Instances 1 public and 3 privates according to VPCs
Connected Public instance with SSH and checked the connectivity of 3 private instances with private IPs which was working fine.

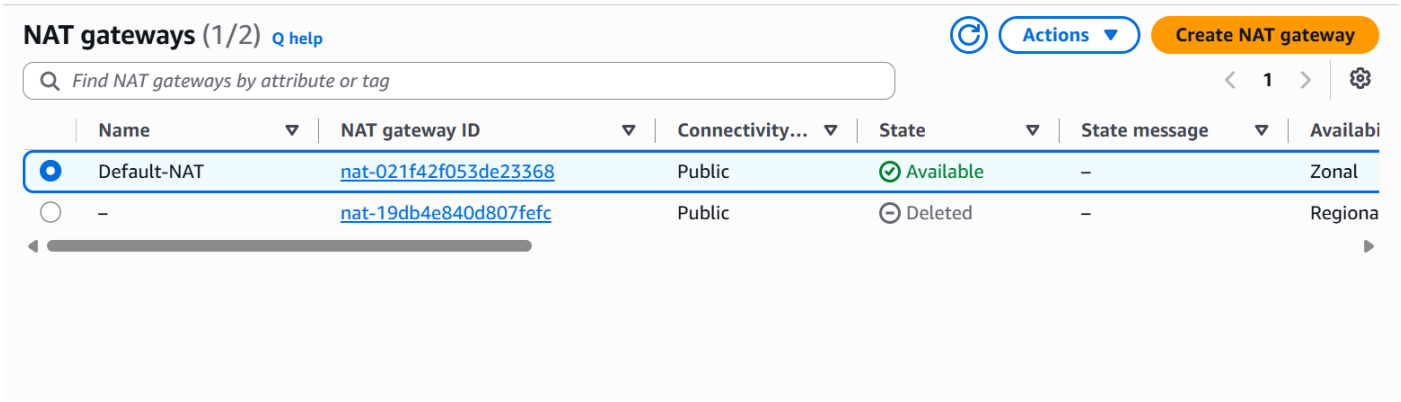
Results Below:

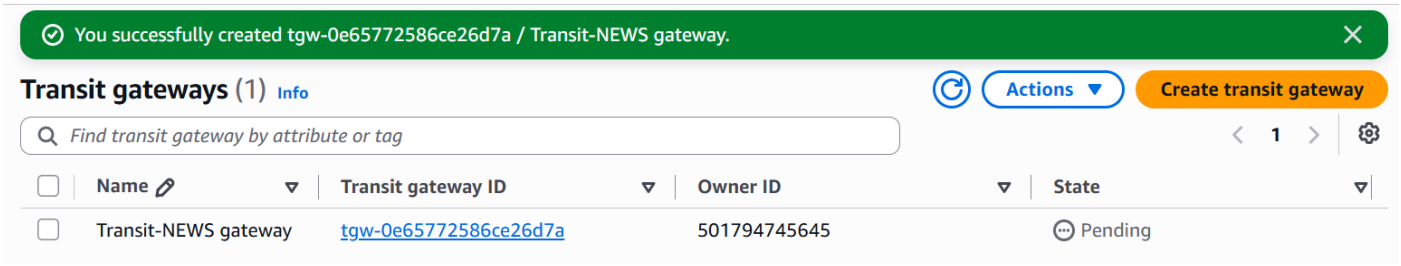


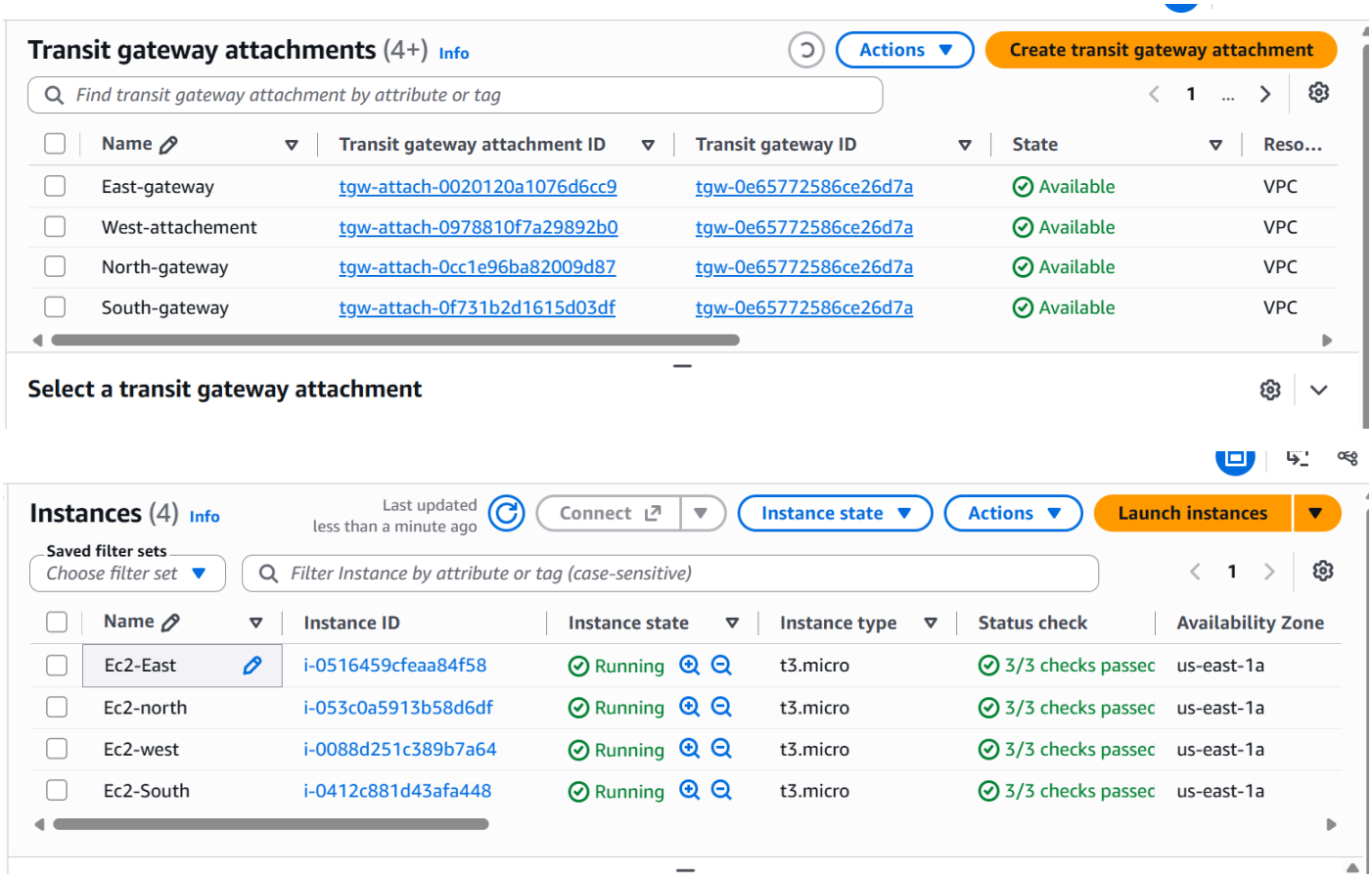





Internet gateways (1/2) Q help
Actions
Create internet gateway
Find internet gateways by attribute or tag
1
Name
Internet gateway ID
State
VPC ID
– igw-04e6fea3554cef1db Attached vpc-0262979be2277efc0
Default-IGW igw-06112cb5a1a57618e Attached vpc-06f6c89f0c190f1a2 | vpc-w


NAT gateways (1/2) Q help
Actions
Create NAT gateway
Find NAT gateways by attribute or tag
1
Name
NAT gateway ID
Connectivity...
State
State message
Availabi
Default-NAT nat-021f42f053de23368 Public Available – Zonal
– nat-19db4e840d807fefc Public Deleted – Regiona


You successfully created tgw-0e65772586ce26d7a / Transit-NEWS gateway.
Transit gateways (1) Info
Actions
Create transit gateway
Find transit gateway by attribute or tag
1
Name
Transit gateway ID
Owner ID
State
Transit-NEWS gateway tgw-0e65772586ce26d7a 501794745645 Pending


Transit gateway attachments (4+) Info
Actions
Create transit gateway attachment
Find transit gateway attachment by attribute or tag
1
Name
Transit gateway attachment ID
Transit gateway ID
State
Reso...
East-gateway tgw-attach-0020120a1076d6cc9 tgw-0e65772586ce26d7a Available VPC
West-attachement tgw-attach-0978810f7a29892b0 tgw-0e65772586ce26d7a Available VPC
North-gateway tgw-attach-0cc1e96ba82009d87 tgw-0e65772586ce26d7a Available VPC
South-gateway tgw-attach-0f731b2d1615d03df tgw-0e65772586ce26d7a Available VPC
Select a transit gateway attachment

Instances (4) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
1
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Ec2-East i-0516459cfeaa84f58 Running t3.micro 3/3 checks passec us-east-1a
Ec2-north i-053c0a5913b58d6df Running t3.micro 3/3 checks passec us-east-1a
Ec2-west i-0088d251c389b7a64 Running t3.micro 3/3 checks passec us-east-1a
Ec2-South i-0412c881d43afa448 Running t3.micro 3/3 checks passec us-east-1a

```
^C
--- 10.10.0.8 ping statistics ---
83 packets transmitted, 25 received, 69.8795% packet loss, time 85262ms
rtt min/avg/max/mdev = 0.545/0.766/3.927/0.701 ms
[ec2-user@ip-10-20-0-6 ~]$ ping 10.40.0.7
PING 10.40.0.7 (10.40.0.7) 56(84) bytes of data.
64 bytes from 10.40.0.7: icmp_seq=1 ttl=126 time=2.99 ms
64 bytes from 10.40.0.7: icmp_seq=2 ttl=126 time=0.464 ms
64 bytes from 10.40.0.7: icmp_seq=3 ttl=126 time=0.471 ms
64 bytes from 10.40.0.7: icmp_seq=4 ttl=126 time=0.491 ms
64 bytes from 10.40.0.7: icmp_seq=5 ttl=126 time=0.491 ms
^C
--- 10.40.0.7 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4146ms
rtt min/avg/max/mdev = 0.464/0.981/2.991/1.004 ms
[ec2-user@ip-10-20-0-6 ~]$ ping 10.30.0.12
PING 10.30.0.12 (10.30.0.12) 56(84) bytes of data.
64 bytes from 10.30.0.12: icmp_seq=47 ttl=126 time=1.46 ms
64 bytes from 10.30.0.12: icmp_seq=48 ttl=126 time=0.532 ms
64 bytes from 10.30.0.12: icmp_seq=49 ttl=126 time=0.497 ms
64 bytes from 10.30.0.12: icmp_seq=50 ttl=126 time=0.513 ms
64 bytes from 10.30.0.12: icmp_seq=51 ttl=126 time=0.536 ms
^C
--- 10.30.0.12 ping statistics ---
51 packets transmitted, 5 received, 90.1961% packet loss, time 52020ms
rtt min/avg/max/mdev = 0.497/0.706/1.455/0.374 ms
[ec2-user@ip-10-20-0-6 ~]$ ping 10.10.0.8
PING 10.10.0.8 (10.10.0.8) 56(84) bytes of data.
64 bytes from 10.10.0.8: icmp_seq=1 ttl=126 time=1.82 ms
64 bytes from 10.10.0.8: icmp_seq=2 ttl=126 time=1.05 ms
64 bytes from 10.10.0.8: icmp_seq=3 ttl=126 time=0.547 ms
64 bytes from 10.10.0.8: icmp_seq=4 ttl=126 time=0.568 ms
64 bytes from 10.10.0.8: icmp_seq=5 ttl=126 time=0.543 ms
64 bytes from 10.10.0.8: icmp_seq=6 ttl=126 time=0.558 ms
^C
--- 10.10.0.8 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5085ms
rtt min/avg/max/mdev = 0.543/0.846/1.818/0.470 ms
[ec2-user@ip-10-20-0-6 ~]$
```

Challenges faced during task:

All Instance security group settings didn't allow to connect even public instance also can't able to connect so changed the all Instances inbound security by giving all traffic my ipv4.

Forgot to add subnet associations to route tables.

Forgot to edit routes and add other 3 instance IPs with transits gateway I was just add 1 transit with IP of 0.0.0.0/16

Forgot to add Routes of other instances.

Configure VPC endpoints to securely access AWS services without internet gateways or NAT gateways, ensuring data privacy and minimizing exposure to external threats

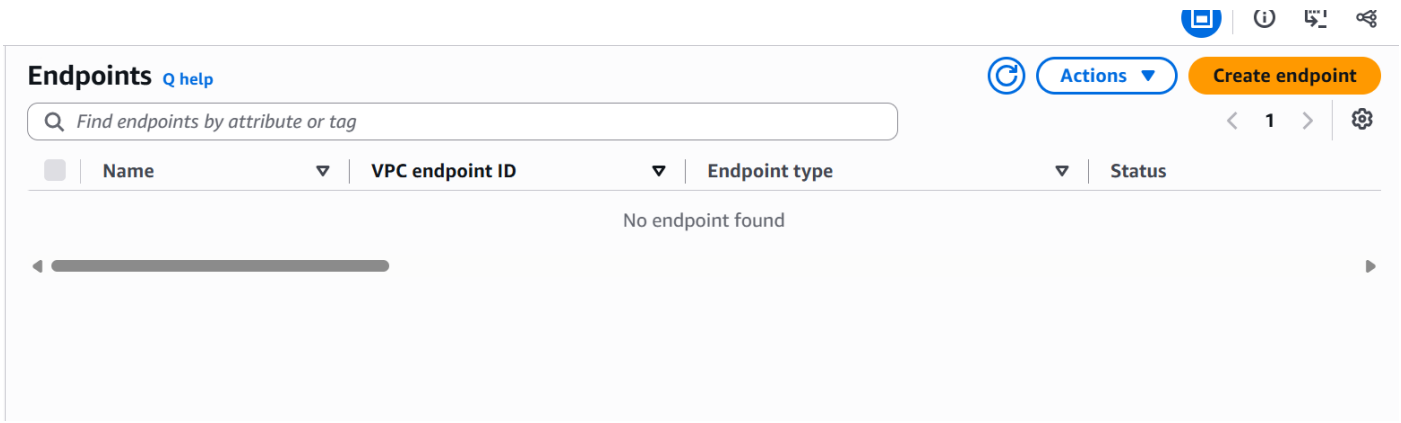
Steps:

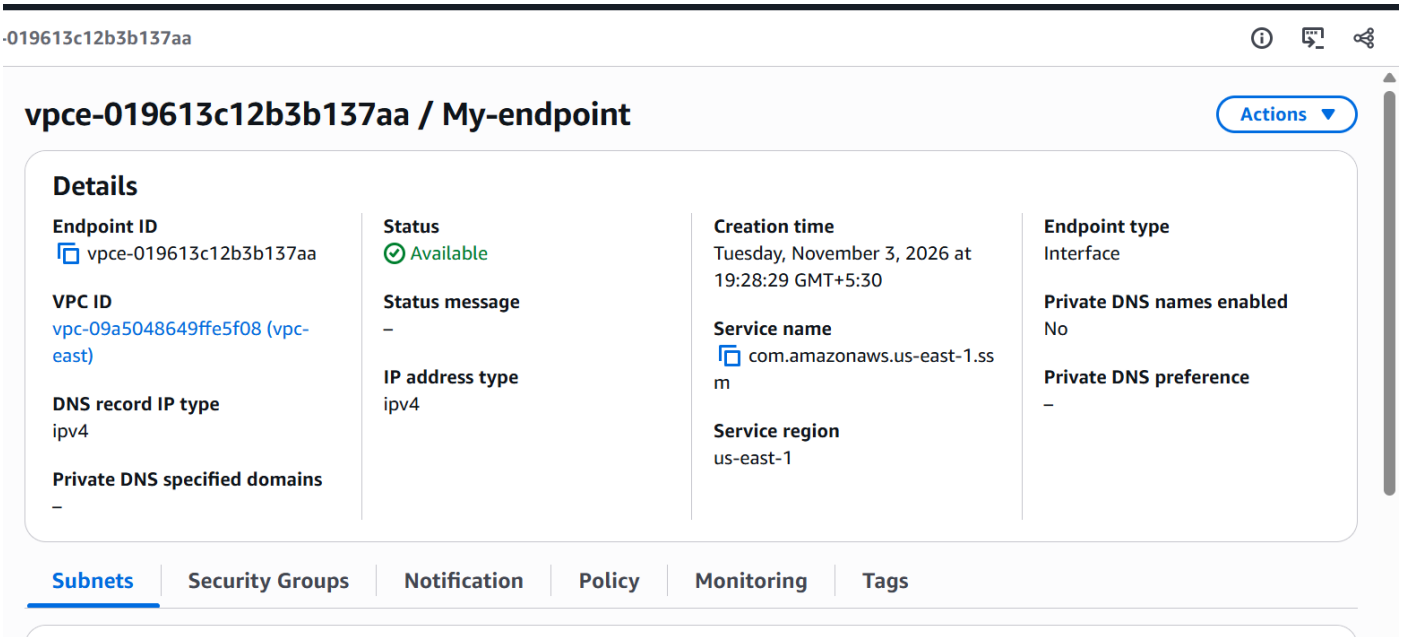
Go to Endpoint and create 1 endpoint → created new security group to allow inbound rules HTTPS and IPv4.

Created s3 bucket → created IAM user → created access key

Login in Public Instance with ssh key → checked AWS version → `aws-cli/2.38.0 Python/3.9.25 Linux/6.18.56-125.164.amzn2023.x86_64 source/x86_64.amzn.2023`
→ AWS configure → `AWS Access Key ID [None]: AKIAXJVKMFUWQYBK43NN`
`AWS Secret Access Key [None]: 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`
`Default region name [None]:`
`Default output format [None]: json`
→ AWS s3 ls

Results Below:





Inbound rules Info

Type Info	Protocol Info	Port range Info	Source type Info	Source Info	Description - optional Info	
HTTPS	TCP	443	Custom	10.10.0.0/16		Delete
HTTPS	TCP	443	Custom	10.20.0.0/16		Delete
HTTPS	TCP	443	Custom	10.30.0.0/16		Delete
HTTP	TCP	80	Custom	10.40.0.0/16		Delete

Add rule

Security groups (1/6) Q help

Filter resources

Group ID	Group name	VPC ID	Description
sg-0debae442d8c239a5	default	vpc-09a5048649ffe5f08	default VPC security group
sg-02757393d4e6fde86	default	vpc-06f6c89f0c190f1a2	default VPC security group
sg-08ab22b85ff71a301	default	vpc-0ec9b15f969744484	default VPC security group
sg-081dc11253f535ffb	default	vpc-09683edb58faddd21	default VPC security group
sg-0e6bfb8d56912a560	default	vpc-0262979be2277efc0	default VPC security group
sg-005cd5021cd1d2382	webserver4endpoint	vpc-0262979be2277efc0	allow endpoint connection

Successfully created bucket "viquar-s3.bucket"

viquar-s3.bucket Info

Objects | Metadata | Properties | Permissions | Metrics | Management | File systems | Access Points

Objects (0)

Copy S3 URI | Copy URL | Download | Open | Delete | Actions | Create folder | Upload

Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more

Find objects by prefix

Name	Type	Last modified	Size	Storage class

No objects

You don't have any objects in this bucket.

Upload

viquar Info

Delete

Summary

ARN
arn:aws:iam::501794745645:user/viquar

Created
November 03, 2026, 19:29 (UTC+05:30)

Console access
Disabled

Last console sign-in
-

Access key 1
Create access key

Permissions | Groups | Tags (1) | Security credentials | Last Accessed

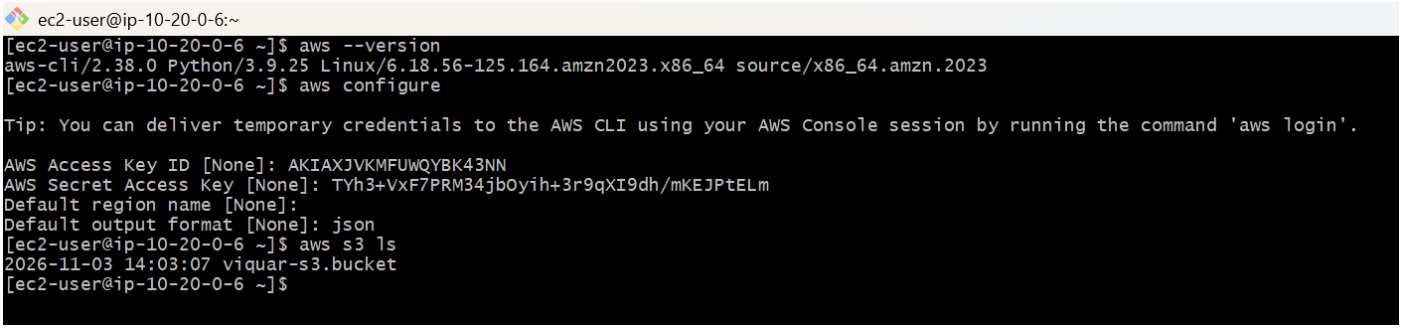
Permissions policies (1)

Simulate | Remove | Add permissions

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type

Search | All types



Challenges faced during task:

While using existing security group it wasn't working so created new security group.